

KneeDINO: A Knee MRI Vision Foundation Model using Self-Supervised Learning

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Abstract. Knee MRI is widely used for diagnosing musculoskeletal abnormalities, but foundation models specifically adapted to knee MRI remain limited. In this work, we proposed KneeDINO, a knee MRI-specialized vision foundation model adapting DINOv3 through domain-specific self-supervised learning. KneeDINO was pretrained on approximately 269,000 unlabeled sagittal PD fat-suppressed knee MRI slices. Compared with DINOv3, KneeDINO achieved improved classification accuracy and DSC across most anatomical structures, demonstrating the effectiveness of knee MRI-specific self-supervised learning.

Keywords: Medical Foundation Model, Self-supervised Learning, Knee MRI.

1 Introduction

Knee MRI is widely used for the diagnosis of musculoskeletal abnormalities. Although general medical foundation models and MRI-specific foundation models have recently gained increasing attention, foundation models specialized for knee MRI remain limited. In this work, we proposed KneeDINO, a knee MRI-specialized vision foundation model, by adapting a DINOv3 through domain-specific self-supervised learning.

2 Method

For self-supervised learning, we used a large-scale unlabeled knee MRI dataset collected at Seoul National University Hospital (SNUH), consisting of approximately 269,000 sagittal PD fat-suppressed knee MRI slices. We proposed KneeDINO, a knee MRI-specialized vision foundation model using DINOv3 [1] as a backbone. As shown in Figure 1a, unlabeled knee MRI slices were transformed into global and local crops

and used for self-supervised teacher-student training. This domain-adaptive pretraining enabled KneeDINO to better capture knee MRI-specific anatomical features.

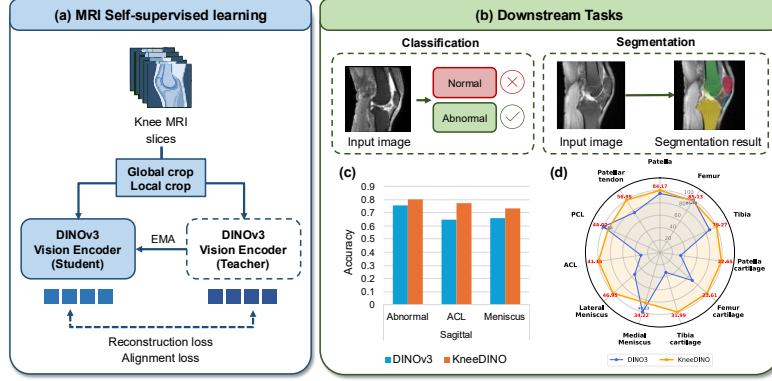


Fig. 1. Overview of the proposed KneeDINO. (a) KneeDINO performs domain-adaptive self-supervised learning on unlabeled knee MRI slices. (b) The pretrained KneeDINO encoder is evaluated on classification and few-shot segmentation tasks. (c) Classification accuracy on an external dataset. (d) Few-shot segmentation performance for knee anatomical structures.

3 Results

As shown in Figure 1b, KneeDINO was evaluated on two downstream tasks: classification and segmentation. For classification, linear probing was performed with the frozen encoder on the MRNet [2] dataset. For segmentation, few-shot experiments were conducted on an internal knee MRI dataset, where five annotated slice images were used for each anatomical class. During segmentation training, the encoder was frozen and only a lightweight linear segmentation head was trained.

As shown in Figure 1c, KneeDINO showed improved classification accuracy over DINOv3 on the sagittal MRNet dataset, achieving 0.800, 0.775, and 0.733 for abnormal exam, ACL tear, and meniscus tear classification, respectively. In addition, Figure 1d demonstrated that KneeDINO achieved higher DSC than DINOv3 across most anatomical structures in few-shot segmentation.

Acknowledgments. This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) [NO.RS-2021-II211343, Artificial Intelligence Graduate School Program (SNU)]

References

1. Siméoni, Oriane, et al. "Dinov3." *arXiv preprint arXiv:2508.10104* (2025)
2. Bien, Nicholas, et al. "Deep-learning-assisted diagnosis for knee magnetic resonance imaging: development and retrospective validation of MRNet." *PLoS medicine* 15.11 (2018)